

Do **not** write solutions on this page.

11. [Maximum mark: 18]

A line L is defined by $L: -\frac{x}{2} + 1 = y + 4 = \frac{z}{3}$.

(a) Find the equation of L , expressing your answer in the form $\mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$, where $\lambda \in \mathbb{R}$. [3]

(b) Determine the minimum distance from the origin O to the line L . [5]

A plane Π is defined by $\Pi: 6x - 3y + 5z = 24$.

(c) Verify that Π contains L . [3]

A second line M is parallel to Π .

The line M passes through the point $(4, 1, 2)$ and intersects the z -axis.

(d) Find the equation of M , expressing your answer in the form $\mathbf{s} = \mathbf{c} + \mu \mathbf{d}$, where $\mu \in \mathbb{R}$. [7]

